Linear Algebra and Analytic Geometry

Unit 1: Matrices and Systems

Worked Solutions to Exercises 1–34

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1. **State, with justification, whether the following statements are true or false.**

   Let $n \in \mathbb{N}^+$ and $A, B, C \in \mathcal{M}_n(\mathbb{K})$.

   a) **For any square matrices $\boldsymbol{A}$ and $\boldsymbol{B}$ of the same order, $\boldsymbol{AB = BA}$.**

   Universal statement:

   $$\forall A, B \in \mathcal{M}_n(\mathbb{K}) : AB = BA.$$

   False. Proof by counterexample.

   Consider

   $$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

   Then,

   $$AB = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \neq \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = BA.$$

   Therefore, the statement is false.

   Sufficient condition:

   If A and B are diagonal matrices, then, for all $i, j \in \{1, \ldots, n\}$,

   $$(AB)_{ii} = a_{ii}b_{ii} = b_{ii}a_{ii} = (BA)_{ii},$$
   $$(AB)_{ij} = 0 = (BA)_{ij}, \quad i \neq j.$$

   Hence, all corresponding entries are equal, so

   $$\boxed{AB = BA}.$$

b) **For any square matrices A and B of the same order, if $AB = 0$, then $A = 0$ or $B = 0$.**

Universal statement:

$$\forall A, B \in \mathcal{M}_n(\mathbb{K}) : AB = 0_n \implies (A = 0_n \vee B = 0_n).$$

False. Proof by counterexample.

Consider

$$A = B = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \neq 0_2.$$

However,

$$AB = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = 0_2$$

Thus,

$$AB = 0_2 \quad \wedge \quad A \neq 0_2 \quad \wedge \quad B \neq 0_2.$$

Therefore, the statement is false.

Sufficient condition:

If either A or B is invertible, the implication holds.

Case 1: A is invertible.

Multiplying by A^{-1} on the left:

$$\begin{aligned} AB = 0_n &\implies A^{-1}(AB) = A^{-1}0_n \\ &\implies (A^{-1}A)B = 0_n \\ &\implies I_n B = 0_n \\ &\implies \boxed{B = 0_n}. \end{aligned}$$

Case 2: B is invertible.

Multiplying by B^{-1} on the right:

$$\begin{aligned} AB = 0_n &\implies (AB)B^{-1} = 0_n B^{-1} \\ &\implies A(BB^{-1}) = 0_n \\ &\implies AI_n = 0_n \\ &\implies \boxed{A = 0_n}. \end{aligned}$$

Hence, if at least one factor is invertible, the other must be zero whenever $AB = 0_n$.

c) **For any square matrices A, B, and C of the same order, if $AB = AC$, then $B = C$.**

Universal statement:

$$\forall A, B, C \in \mathcal{M}_n(\mathbb{K}) : AB = AC \implies B = C.$$

False. Proof by counterexample.
Consider

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad C = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}.$$

Then,

$$AB = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = AC,$$

while $B \neq C$.
Therefore, the statement is false.

Sufficient condition:
If A is invertible, there exists $A^{-1} \in \mathcal{M}_n(\mathbb{K})$ such that

$$A^{-1}A = AA^{-1} = I_n.$$

Multiplying both sides of $AB = AC$ by A^{-1} on the left:

$$\begin{aligned} AB = AC &\implies A^{-1}(AB) = A^{-1}(AC) \\ &\implies (A^{-1}A)B = (A^{-1}A)C \\ &\implies I_nB = I_nC \\ &\implies \boxed{B = C}. \end{aligned}$$

Hence,

$$\boxed{A \text{ invertible} \wedge AB = AC \implies B = C.}$$

d) **If, for a square matrix $\boldsymbol{A}$, there exists $\boldsymbol{k} \in \mathbb{N}$ such that $\boldsymbol{A^k = 0}$, then $\boldsymbol{A = 0}$.**
Universal statement:

$$\forall A \in \mathcal{M}_n(\mathbb{K}) : \left(\exists k \in \mathbb{N}^+ : A^k = 0_n\right) \implies A = 0_n.$$

False. Proof by counterexample.
Consider

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \neq 0_2.$$

For $k = 2$,

$$A^2 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = 0_2$$

Thus,

$$\exists k = 2 \in \mathbb{N}^+ : A^k = 0_2, \quad A \neq 0_2.$$

Therefore, the statement is false.

Sufficient condition:
Suppose A is idempotent, i.e.,

$$A^2 = A.$$

By mathematical induction,

$$\forall p \in \mathbb{N}^+ : A^p = A.$$

Indeed, the base case $A^1 = A$ holds. Assuming $A^p = A$ for some $p \geq 1$, multiplying by A gives

$$\begin{aligned} A^{p+1} &= A^p A \\ &= AA \\ &= A^2 \\ &= A. \end{aligned}$$

Now, if

$$\exists k \in \mathbb{N}^+ : A^k = 0_n,$$

then, since $A^k = A$,

$$A = A^k = 0_n.$$

Therefore,

$$\boxed{A = 0_n}.$$

A matrix that is both idempotent and nilpotent must be zero.

e) **For any square matrices A and B of the same order, $(AB)^2 = A^2B^2$.**
Universal statement:

$$\forall A, B \in \mathcal{M}_n(\mathbb{K}) : (AB)^2 = A^2B^2.$$

False. Proof by counterexample.
Consider

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}.$$

Computing AB,

$$AB = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}.$$

Therefore,

$$(AB)^2 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}.$$

However,

$$A^2 = 0_2, \qquad B^2 = 0_2,$$

so

$$A^2B^2 = 0_2.$$

Hence,

$$(AB)^2 \neq A^2B^2,$$

and the statement is false.

Sufficient condition:
Suppose $AB = BA$.
Then,

$$\begin{aligned} (AB)^2 &= (AB)(AB) \\ &= A(BA)B \\ &= A(AB)B \\ &= (AA)(BB) \\ &= \boxed{A^2B^2}. \end{aligned}$$

Therefore,

$$\boxed{AB = BA \implies (AB)^2 = A^2B^2.}$$

2. Final item. Remove when done.